

Using Spatial Data to Evaluate Employment Effects of Seattle's Minimum Wage Increase

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1 Abstract

This paper investigates the employment effects of Seattle’s 2015 Minimum Wage Ordinance using spatial data from the LEHD Origin-Destination Employment Statistics (LODES). By focusing on the flow of commuters into Seattle before and after the policy’s implementation, I estimate the elasticity of labor inflows with respect to the relative minimum wage ratio between Seattle and surrounding regions. The analysis identifies statistically significant negative commuting elasticities: -0.22 for all workers and -0.63 for low-wage workers earning under \$1250/month, suggesting a meaningful decline in cross-border employment, particularly among low-wage commuters.

These findings contribute to a growing body of literature that recognizes nuanced spatial and sectoral effects of local minimum wage increases. While aggregate job counts may remain stable, this study highlights that employment opportunities can shift across municipal boundaries in response to wage policies. The results align with prior research indicating that minimum wage hikes can reduce hiring and hours worked, especially for less-experienced workers, and point toward spillover effects that reshape regional labor markets.

2 Introduction

Increases in minimum wage may have negative employment effects. Using spatial data, this paper analyzes employment trends following a city-level minimum wage policy passed in 2015.

A considerable body of work exists on minimum wage unemployment effects, including research on the Seattle Minimum Wage Ordinance in particular. Prior literature is mixed on whether or not increasing the minimum wage has negative labor market effects.

Reich et al. 2017 finds positive effects of the 2015 Minimum Wage Ordinance on wages in food service through a synthetic-control analysis of county and city-level data for 2009 to 2016 (control generated elsewhere in WA and the U.S. in whole). Their research, under the banner of the Center on Wage and Employment Dynamics, also concludes that there is no decrease in employment (Reich et al. 2017). Their finding of no decrease in unemployment reinforces Card and Krueger’s famous, foundational finding from their difference-in-differences study of New Jersey and Pennsylvania fast

food restaurants, finding no decrease in employment after a minimum wage increase (Card et al. 1993). However, many economists persist in the notion that raising the minimum wage is a tradeoff of higher wages for some versus job losses for others. In Neumark’s 2013 analysis of newer studies aiming to account for spatial heterogeneity, they reject a lack of disemployment effects and cite teen employment elasticities near -0.3 (Neumark et al. 2013). The debate on minimum wage and disemployment is as "old as the Department of Labor," says Neumark. Even the handful of studies completed on Seattle’s Minimum Wage Ordinance itself, since 2015, have found conflicting results. Contra Reich, Jardim et al. 2022, using synthetic controls generated from elsewhere in WA, found a decline in aggregate employment for relevant employers in terms of hours worked, with “no change in the probability of continued employment but significant reductions in hours, particularly for less experienced workers” (Jardim et al. 2022). Jardim’s 2022 study, which utilized administrative microdata to precisely define low-wage workers, also found that turnover declined, as did hiring of new workers into low wage jobs.

Of course, it is possible that aggregate job levels remain the same, but labor market conditions still change for the worse in certain ways. Unemployment or underemployment for commuting workers, for example, may result from the pushing of business out of Seattle, to outlying areas. Supporting this theory is Dharmasankar et al. 2022. They use spatial data to find "strong, statistically significant spillover effects on establishment entries at the census block level in the hospitality and retail sectors 1-2 years after Seattle’s minimum wage is announced [...] implying an increase in the number of establishments entering the areas surrounding Seattle after the announcement of the minimum wage."

To contribute to this discussion, and further explore this question, this paper leverages a novel dataset to further examine whether or not unemployment effects occur. I track relevant changes in the flow of workers into the city after the 2015 increase using the LEHD Origin-Destination Employment Statistics (LODES) dataset, which reports earnings and employment, by employment sector, for home/work census-block pairs. Here, the use of spatial data allows for specificity about individual-level labor patterns associated with the policy change of interest.

After basic tests of aggregate trends, I conduct a panel data analysis on Y_1 , the count of workers living outside of city limits but working within city limits, to identify the inflow of commuters into the Seattle municipal area

after the 2015 increase. I estimate the elasticity of labor flow, or commuting elasticity, with respect to changes in the minimum wage ratio between Seattle and surrounding regions of the state. I find a commuting elasticity of -0.22 for all workers and a commuting elasticity of -0.63 for low-wage workers (earning less than \$1250 a month). The statistically significant negative effects suggest that workers outside of Seattle, especially low-wage workers, migrate away from working jobs within the city limits. These results are consistent with the negative effects found by the county-level analysis of Todd et al. 2022, which finds a commuting elasticity of -1.29 for workers within 11km and -0.45 for workers within 22km due to local minimum wage increases. This is also consistent with cross-state analysis of commuting patterns of low-wage workers, which suggest that workers commute away from minimum wage-increases rather than towards them (McKinnish et al. 2017).

3 Background

In June 2014, the city of Seattle, Washington passed unprecedented legislation to begin to raise wages for workers in the city to \$15 per hour. The legislation, known as Seattle’s Minimum Wage Ordinance (MWO), went into effect on April 1, 2015. It specified gradual increases of the city’s minimum wage to \$15/hour, based on the size of the employing firm. After reaching \$15/hour, the minimum wage has increased every year on January 1 based on the Consumer Price Index (CPI) for the Seattle Tacoma Bremerton area. Seattle now has among the highest minimum wage of any major American city, at nearly \$21 an hour. The Minimum Wage Ordinance was intended to help working-class Seattleites earn and save, reducing wealth inequality in an extremely unequal country. However, neoclassical economic theory suggests there are long-run problems with policies like the Minimum Wage Ordinance. In particular, imposing a price floor on the labor market should create deadweight-loss, as the equilibrium wage (e.g. \$10) where labor supply meets labor demand exists below that floor (e.g. \$15) (Mankiw, Principles of Economics, 8th Edition). In theory, employment should decrease after legislation like the MWO, decreasing turnover and the rate of new hires into low-wage jobs, as labor demand can no longer meet labor supply.

4 Data and Empirical Strategy

My analysis uses LODES occupation count data for Washington State from 2002 to 2022. LODES dataset variables include the number of jobs in a given work census block for the North American Industry Classification System (NAICS) code for Accommodation and Food Services), as well as Origin-Destination information describing pairs of home-work geocodes. Importantly, LODES integrates state administrative data on annual earnings from unemployment insurance. LODES also reports detailed home and work geocode data on the census-block-pair level. LODES data for Washington State contains over 2 million annual observations of worker origins and destinations. By way of comparison, Reich et al.’s analysis uses the Bureau of Labor and Statistics’ Quarterly Census of Employment and Wages (QCEW), which is reported on the employer level. The greater specificity of this analysis will hopefully shed light on labor market patterns surrounding the Minimum Wage Ordinance.

I estimate counts of workers living outside of city limits, but working within city limits, by utilizing the R statistical programming language in conjunction with geospatial software QGIS. I used a shapefile from the City of Seattle via geospatial data host service Koordinates to draw the municipal limits. I intersected this boundary with the annual LODES data to find job flows into Seattle by filtering for pairs with work geocodes within, and home geocodes without, this boundary.

The key regression is intended to identify changes in flows as Seattle’s MWO gradually takes effect. The unit of observation is (home-work census-block geocode-pair)-(years). Each of these pairs, or flows, represents the commute of minimum of 1 to a maximum 126 Washington State residents, from where they live outside of Seattle, into the Seattle city limits. For my primary parameter of interest, following Todd et al. 2022, I estimate the census-block-level elasticity of labor flows (commuters from outside city limits to within city limits) on the relative minimum wage ratio of Seattle to its neighboring areas (NBER). Our parameter of interest is β_1 in the following equation:

$$\log(y_{hwt}) = \beta_0 + \beta_1 \log(MW_1/MW_0) + \delta_t + \epsilon_{hwt} \quad (1)$$

Where y_{hwt} is the log of commuters into Seattle for time t from residence census tract h into work census tract w . We control for potential unobserv-

ables by time effects δ_t . The coefficient β_1 , the primary parameter of interest, gives the elasticity of labor flows with respect to the gradual increase in the relative minimum wage ratio MW1/MW0.

For Table 1, I filtered for census-block-pairs where minimum data for all jobs were present for all 9 years from 2011 to 2019. For Table 2, For Table 1, I filtered for census-block-pairs where minimum data for low-wage (<\$1250 a month) jobs were present for all 9 years from 2011 to 2019.

As seen in the regression, the gradual increase in the Seattle minimum wage, where surrounding areas do not increase in minimum wage, allows us to estimate the impact of a higher minimum wage on job flows. For our independent variable, I use the ratio between the small firm (less than 500 employees) minimum wage and the rest of the state minimum wage. This ratio is 1 before 2015 and greater than 1 afterward (e.g. 1.27 in 2016). This ratio, which would be higher when applied to large firms, is our proxy for the minimum wage differential between a given home geography and a given work geography. I use data from the Washington State Department of Labor & Industries to construct the minimum wage ratio from 2011 to 2019.

A key concern with this minimum wage ratio proxy approach is that other localities around Seattle, such as Everett, may have raised their minimum wages in tandem, thus nullifying the differential with the rest of the state. SeaTac raised its minimum wage to \$15 in 2013, two years before the Seattle policy, Tacoma raised its minimum wage to \$16 in 2015, and Tukwila raised its minimum wage to \$15 in 2022. In an ideal setting, my regression would include minimum-wage ratio data based off of the residence geocode, not a blanket ratio for the rest of the state. However, granular information on municipal boundaries and timing of local minimum-wage ordinances is difficult to find. Seattle’s minimum-wage ordinance was among the first in the state; SeaTac and Tacoma are among only very few localities that one should expect to have similar raises at the same time.

Before conducting my regression analysis, I conducted basic aggregate tests on the overall trends in people commuting into Seattle after 2015. Using the online LEHD OnTheMap tool, LODES count data was filtered for jobs within Seattle filled with people living outside of Seattle from 2003-2022.

At a first approximation, as seen in Figure 1, there is no obvious decrease in the count of commuting jobs held after the 2015 increase in the minimum wage. To validate the shapefile and LODES data, I replicated the OntheMap aggregate trends in Figure 1 by filtering the LODES data in R for individuals with work geocodes within the Seattle city boundary shapefile and home

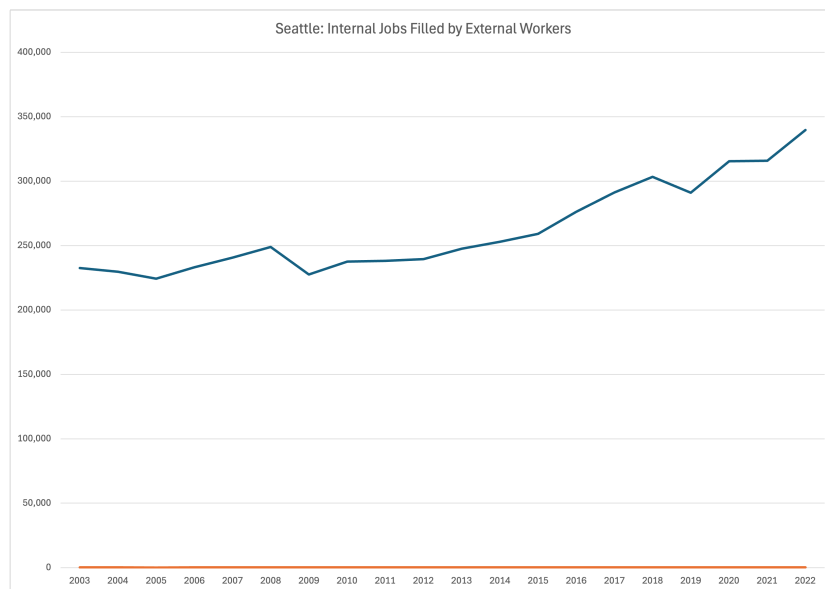


Figure 1: Private Primary Jobs within Seattle held by External Workers.

geocodes outside the boundaries, for 2013-2017.

These total counts match employment trends from the OnTheMap tool, though they are approximately 10% higher, likely due to including nonprimary (when people have jobs that are not their primary means of income) and nonprivate jobs.

The LODS data contains information on whether or not a given job was within the NAICS-coded Accommodation and Food Services sector. I conducted two tests on aggregate trends within this data.

First, as you can see in Figure 2, the total number of food-service jobs within Seattle city limits remained relatively stable through 2015, though the COVID-19 pandemic led to a large drop in 2020.

Second, I calculated average annual changes in the count of food-service jobs, per census-block, within the treatment zone. In Table 4, we see that each census block hired 1-2 new workers every year until 2019 and 2020, where we see a slight decrease and a staggering decrease, respectively. The drop in employment in 2020 is likely correlated with the COVID-19 pandemic and its effects on food-service establishments.

At this point, neither total flow of workers into the city nor job counts within the city have demonstrated any obvious changes in employment due to

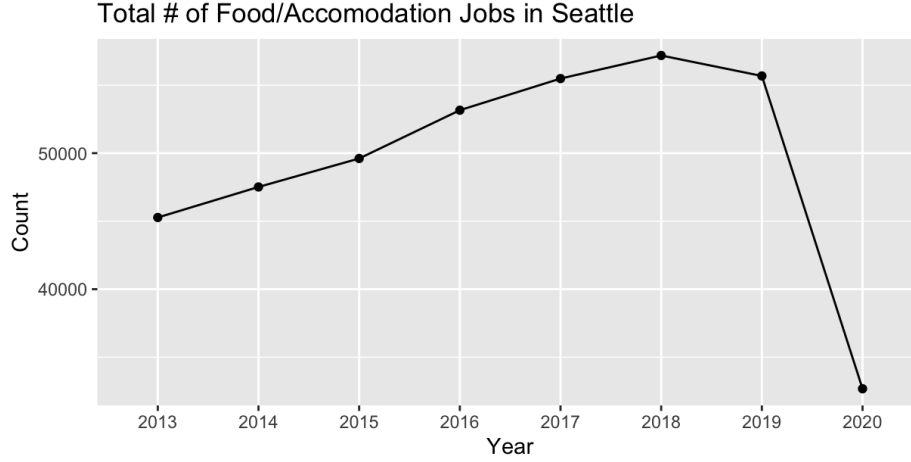


Figure 2: NAICS Accommodation and Food Services Sector Job Counts from LODES.

the 2015 ordinance. However, employment, including of commuters, could be increasing or staying stable for reasons exogenous to the minimum wage increase, including macroeconomic conditions. Census-block level spatial analysis allows one to gain clarity on what is happening on the individual level. Therefore, we would like to investigate across time whether or not people from the same blocks outside of Seattle are staying or leaving the jobs that they have in Seattle, as the minimum-wage differential increases.

5 Results

In Tables 1 and 2, I present the results of the regression outlined in equation (1). In Table 1, we find statistically significant negative commuting elasticities for total job flows with respect to the log minimum wage ratio across 1.8 million commuting-pair-year observations over nine years (2011-2019). In Table 2, we find greater, and also statistically significant, negative commuting elasticity for low-wage (<\$1250 a month) job flows across 0.27 million.

We find minimal standard errors and $p < 0.01$ for both coefficients referring to the log minimum wage ratio (commuting elasticity).

Our covariates in both Table 1 and 2 show a significant negative fixed effect associated with 2015. This is possibly due to my reporting of the relative minimum wage ratio of 1.00 for 2015, as the minimum wage ordinance

passed in the middle of that year (April). Employers may have made immediate employment changes due to the Minimum Wage Ordinance’s passage that were not reflected in the actual minimum wage ratio until April of 2015. Additionally, the Seattle City Council voted to increase the minimum wage to \$15 as early as June of 2014, so employment decisions may have been made before 2015 (BBC 2015). The presence of this significant negative fixed effect coefficient (in Table 2, this is -12.5% for 2015) should imply that the Minimum Wage Ordinance did indeed have reverberating effects across Seattle’s Labor Market.

The commuting elasticities are -0.215 and -0.633 for all jobs and low-wage jobs, respectively. One basic interpretation of the spatial analysis as modelled by linear regression is that a 20% increase in the relative minimum wage ratio would result in a 4% decrease in total job flows into Seattle, and a 13% decrease in low-wage job flows into Seattle. Overall, this suggests that less people commute into the city once the minimum wage is raised there.

6 Conclusions/Discussion

The negative results of my primary regression on LODES spatial data, combined with null results of the aggregate trend analysis, are well situated in the body of minimum-wage employment literature that finds negative effects on job flows, but not on aggregate job counts. For comparison, the commuting elasticity methodology follows Todd et al. 2022, who use LODES data to estimate for 1181 unique county pairs, where one of two bordering counties experience minimum-wage increases. They find negative commuting elasticity of -1.29 for workers within 11km and -0.45 for workers within 22km of local minimum wage increases. This is also consistent with cross-state analysis of commuting patterns of low-wage workers, which suggest that workers commute away from minimum wage-increases rather than towards them (McKinnish et al. 2017).

The most exciting interpretation of these negative commuting elasticities concerns the spatial spillover effect posited by Dharmasankar et al. 2022. If people are commuting into Seattle less as the minimum wage ratio increases, it is plausible that the Minimum Wage Ordinance pushed business out of Seattle into the outlying regions. I believe this is supported well by the results in Tables 1 and 2. Dharmasankar et al. 2022 find a corresponding decline in hospitality and retail sector entries within Seattle itself, related to the 2015

Minimum Wage Ordinance. Similarly, Jardim et al. 2022 observe that "experienced low-wage workers, tracked longitudinally, exhibit some tendency to shift their work efforts outside of the city." While I found negative results for both total jobs and low-wage jobs, Table 2 of my results demonstrates a particularly high negative commuting elasticity (-0.63) for low-wage workers, making less than \$1250 a month: corresponding to lower-paying retail or hospitality jobs.

Importantly, these results will not necessarily generalize to other localities or to federal minimum wage increases. The minimum-wage disemployment debate is a long and complex one, and this paper concerns a narrow spatial investigation of job flows into Seattle. Also, it is fair to note that the blanket application of the Minimum Wage Ratio to the regression in Equation (1) is a major issue that I would rectify given more time, and insight into the legislative history of Seattle's surrounding localities (e.g. Bellingham, Tacoma, Burien, SeaTac). Further continuation of this research would match residence-geocode-specific relative minimum wage ratios to all commuters into Seattle. Another potential issue with this analysis is the use of synthetic data within at the census-block-level of LODES for privacy reasons. Addressing this issue accurately, beyond the filters described in Section 3, is beyond the scope of this project.

Table 1:

	<i>Dependent variable:</i>
	Log Job Flows
Log Minimum Wage Ratio	−0.215*** (0.005)
t = 2012	−0.002** (0.001)
t = 2013	0.002** (0.001)
t = 2014	0.004*** (0.001)
t = 2015	−0.041*** (0.001)
t = 2016	−0.003*** (0.001)
t = 2017	0.019*** (0.001)
t = 2018	0.005*** (0.001)
Constant	0.107*** (0.001)
Observations	1,784,174
R ²	0.004
Adjusted R ²	0.004
Residual Std. Error	0.291 (df = 1784165)
F Statistic	937.695*** (df = 8; 1784165)
<i>Note:</i>	₁₁ *p<0.1; **p<0.05; ***p<0.01

Table 2:

	<i>Dependent variable:</i>
	Log Low-Wage Job Flows
Log Minimum Wage Ratio	−0.633*** (0.013)
t = 2012	−0.004 (0.002)
t = 2013	−0.003 (0.002)
t = 2014	−0.0002 (0.002)
t = 2015	−0.125*** (0.003)
t = 2016	−0.029*** (0.002)
t = 2017	0.026*** (0.003)
t = 2018	−0.021*** (0.003)
Constant	0.138*** (0.002)
Observations	270,130
R ²	0.037
Adjusted R ²	0.037
Residual Std. Error	0.314 (df = 270121)
F Statistic	1,301.064*** (df = 8; 270121)

Note: ₁₂*p<0.1; **p<0.05; ***p<0.01
Low-wage job above is defined as < \$1250/month.

Table 3: Minimum Wage Ratio (MW_1/MW_0)

	Year	Ratio
1	2011	1.00
2	2012	1.00
3	2013	1.00
4	2014	1.00
5	2015	1.00
6	2016	1.16
7	2017	1.27
8	2018	1.18
9	2019	1.22

Table 4: Average Annual Change in Food-Service Jobs By Census Block

Year	Job Count within Treatment Zone	Mean Change	Standard Deviation	Range
2013	45,271.00	0.00	0.00	0
2014	47,515.00	2.15	27.74	[-153,711]
2015	49,609.00	1.71	24.55	[-350,253]
2016	53,159.00	2.62	24.51	[-295,267]
2017	55,485.00	2.01	21.89	[-211,372]
2018	57,182.00	0.75	25.29	[-240,305]
2019	55,677.00	-1.96	20.28	[-252,159]
2020	32,681.00	-19.34	45.31	[-765,67]

7 References

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